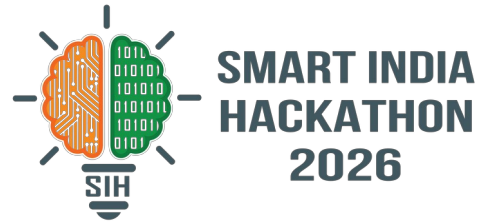
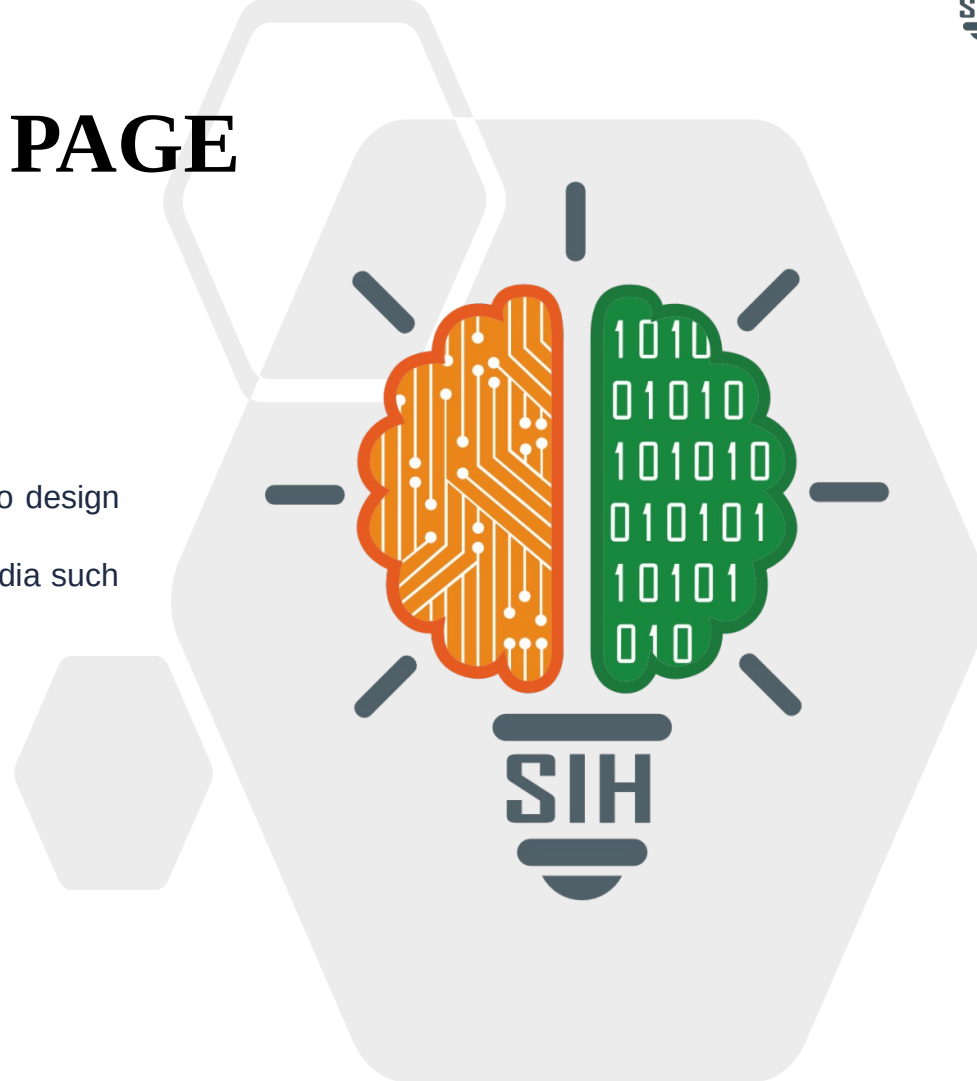


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26201
- **Problem Statement Title-** Student Innovation-There is a need to design drones and robots that can solve some of the pressing challenges of India such as handling medical emergencies, search and rescue operations, etc.
- **Theme-** Robotics and Drones
- **PS Category-** Software
- **Team ID-** 129300
- **Team Name (Registered on portal)** - @TRINETRA



AVIAN: Autonomous Bridge-Inspection Drone

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

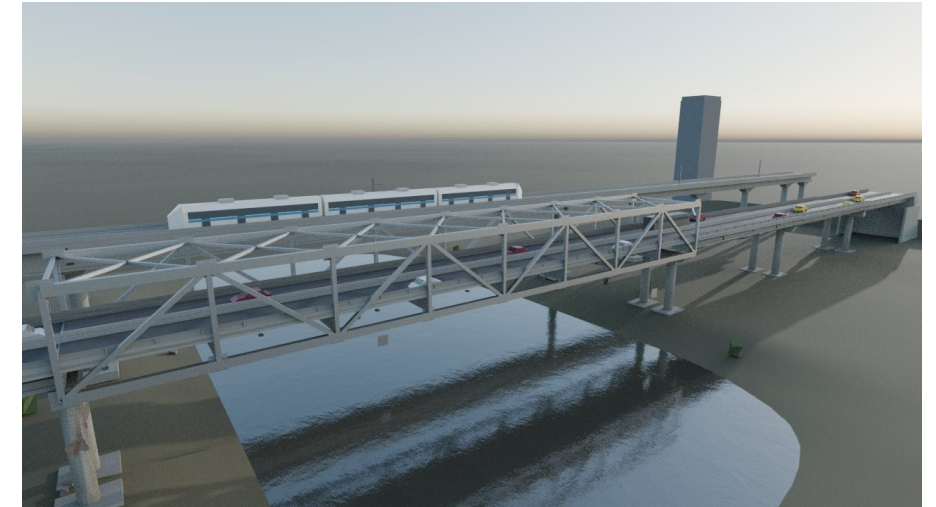
- Plans its own inspection route from the bridge's 3D model — no list of known defects needed
- Flies the route autonomously, steering clear of steel and concrete using on-board range sensing
- A zoom camera captures millimetre-level detail from a safe 8 m standoff
- AI flags cracks, spalling and corrosion and tags each one to the exact bridge member

How it addresses the problem

- India's railways alone have 1,47,523 bridges; 37,689 are over 100 years old. They are checked “by visual perception” with ~40% of bridge-staff posts vacant
- Every span gets a repeatable, geo-tagged image record instead of one person's judgement
- Reaches under-deck and truss zones with no rope access, scaffolding or lane closure

Innovation and uniqueness of the solution

- Route planned from geometry only — code-audited to read zero defect data
- “Survey + zoom” instead of flying close: keeps ≥ 3 m from structure, as China's 2026 UAV bridge-inspection guideline requires
- Every run is scored against a digital twin with 192 known defects, so results are measured, not claimed



Our digital twin: 360 m road bridge + steel truss + metro viaduct, with 192 measured defects

Why now — 9 July 2025

The 40-year-old Gambhira bridge linking Vadodara and Anand districts collapsed into the Mahisagar river, killing 22. Residents had reportedly warned about its condition for years; four engineers were suspended for lapses in maintenance and supervision. (ref. 2)

Technologies to be used (e.g. programming languages, frameworks, hardware)

Digital twin & simulation

- Blender: bridge model, 192 measured defects
- PyBullet: flight physics
- Gazebo + ROS 2 Jazzy: exported world

Autonomy (Python)

- Cascade PID flight controller
- Coverage planner (greedy set-cover)
- Ray-cast range sensor for avoidance

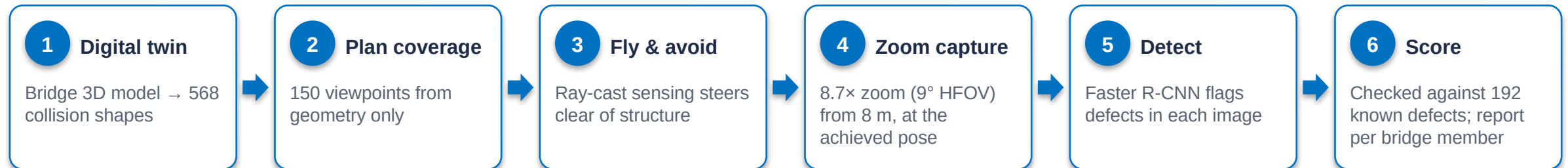
AI defect detection

- PyTorch + torchvision Faster R-CNN (MobileNetV3)
- BSD-3 licence — free to hand over
- Trained on rendered defect images

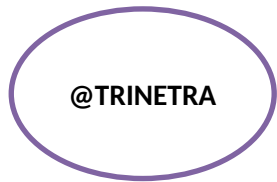
Target hardware (planned)

- PX4 quadcopter + zoom gimbal camera
- LiDAR / visual-inertial SLAM
- Jetson-class on-board computer

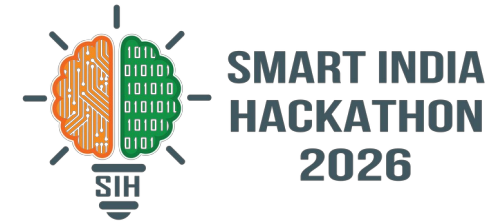
Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype today: all six steps run end-to-end in simulation, with measured results on the next slide. **Next:** run the same stack live in Gazebo + ROS 2, then on PX4 hardware.



FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

Measured in our simulation — not estimated

—	bridge surface coverage — route planned with no defect data (pending freeze)
—	known defects fell inside a settled camera view (pending freeze)
—	flight-log entries show avoidance driven by the sensor, not a map (pending freeze)
—	waypoints with a collision or stall, before vs after sensed avoidance (pending freeze)
0.34	detector mAP@0.5 on 24 held-out defects (synthetic images)

Potential challenges and risks	Strategies for overcoming these challenges
Detection accuracy is the weakest link: 0 detections on 1,000 real bridge photos (precision = recall = 0) — doesn't yet transfer from simulation; on its own mission imagery, unseen-defect recall is 1 of 3 in-scope cases (5 of 20 for all defects, optimistic); false alarms fell from 37.6 to 0.08 per 100 tiles (v1 → v2); loose bolts (~10 px) are below reliable detection at current resolution	Retrain on real bridge datasets (CODEBRIM, dacl10k) on a GPU; higher-resolution capture for bolts; an engineer confirms every flag before it is reported
Collisions in the dense steel truss: stuck/collision rate before → after the Section 4 fix is pending the freeze — see the stat at left	Clearance checks along the whole path, not just at waypoints; slower close-range legs; target under 5% before any field flight
No GPS under the deck; magnetic noise near 25 kV railway overhead lines	Visual-inertial / LiDAR SLAM, as used by Flyability's Elios 3; compass-free heading

Regulatory path: Drone Rules 2021 R&D exemption at prototype stage → DGCA type certificate through a QCI-approved body (~₹1.5 lakh at NTH) → railway safety permits for any work near live 25 kV lines.

IMPACT AND BENEFITS

1,47,523

railway bridges in India

37,689

of them over 100 years old

≈ 33

bridges per bridge-staff member in post (1,47,523 ÷ 4,517)

Source: Standing Committee on Railways, Maintenance of Bridges in Indian Railways (Jan 2019), ref. 1. Inspections rely on the “visual perception” of the inspecting official.

Potential impact on the target audience

- **Indian Railways** — fits the Railway Board's drone-inspection circulars and its Bridge Management System (4D BrIM)
- **NHAI, MoRTH and state PWDs** — e.g. Gujarat's Roads & Buildings department after the Gambhira collapse
- **Metro operators (DMRC)** — viaduct inspection without closing lines
- **Rescue teams** — a quick structural check of a flood- or quake-damaged bridge before convoys cross

Benefits of the solution (social, economic, environmental, etc.)

Social

Earlier warning of failing spans; inspectors off ropes and scaffolds

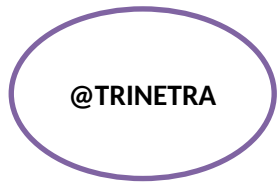
Economic

No lane closures or snoopers trucks; each year's images map onto the same twin, so every defect is tracked over time

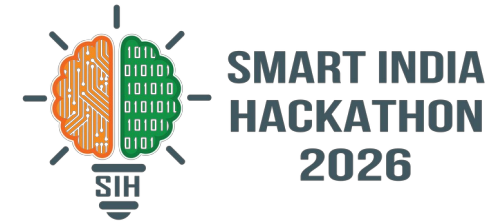
Environmental

Battery-electric; fewer heavy access vehicles and fewer emergency rebuilds

Where it fits: between satellite radar, which tracks millimetre movement across whole regions but can't see under a deck, and manual close inspection, which can but is slow, risky and subjective.



RESEARCH AND REFERENCES



Details / Links of the reference and research work

The problem in India

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Technical basis

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